

Linear Algebra 18.6 - Lecture 4

Factorization into $A = LU$

- Inverse of AB , A^T
- Product of elimination matrices

$$\underline{A = LU} \quad (\text{no more exchange})$$

$$A \cdot A^{-1} = I = A^{-1} \cdot A$$

$$(AB)(B^{-1}A^{-1}) = I$$

do first
do second

$$B^{-1}A^{-1}AB = I$$

Transpose

↳ for square invertible matrices

$$AA^{-1} = I$$

$$(A^{-1})^T A^T = I$$

↳ What is inverse of $A^T \Rightarrow (A^T)^{-1}$
inverse of

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} \xRightarrow{E_{21}} \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \quad \left. \begin{array}{l} \text{Want} \\ A = LU \end{array} \right\}$$

$$A = L \cdot U$$

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

↳ Inverse of $E_{21} \Rightarrow (E_{21})^{-1}$

$L \rightarrow$ lower triangular
 $U \rightarrow$ upper triangular

$$= \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}$$

• Three-by-Three for $A = LU$

$E_{32} E_{31} E_{21} A = U$ (no row exchanges)

$$A = \frac{E_{21}^{-1} E_{31}^{-1} E_{32}^{-1}}{E} U = LU$$

$$= E^{-1} U = LU$$

$$E_{21} E_{31} E_{32}$$

$$E$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix} = E \left(\begin{matrix} \text{left of} \\ A \end{matrix} \right) \quad \underline{E^{-1} U}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = L \left(\begin{matrix} \text{left of} \\ U \end{matrix} \right) \quad \underline{A = LU}$$

$A = LU$

↳ If no row exchanges, the multipliers go directly into L .

• How many operations on $n \times n$ matrix A ?

↳ e.g. 1000×1000 matrix. $\Rightarrow$ How long to solve for inverse?

$n=1000$ $\begin{bmatrix} \dots & \dots & \dots \\ \vdots & \vdots & \vdots \end{bmatrix} \rightarrow \begin{bmatrix} \dots & \dots & \dots \\ 0 & \dots & \dots \\ \vdots & \vdots & \vdots \end{bmatrix} \Rightarrow$ about 100^2 operations.
[for step 1]

elimination
↳ $\begin{bmatrix} \dots & \dots & \dots \\ 0 & \dots & \dots \\ \vdots & \vdots & \vdots \end{bmatrix} \Rightarrow$ about 99^2 operations

• # operations : $n^2 + (n-1)^2 + (n-2)^2 + \dots + 1^2$
(roughly) $\approx \frac{1}{3}n^3$

$\rightarrow$ But this is all based on us ignoring row exchanges

→

• Permutations

↳ What if we take row exchanges into account?

Permutation matrices → Can row exchange

Ex: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

↳ I

Row exchange based on identity matrix.

↳ 6x permutation matrices

↳ Multiply one P by another and the result is a different P.
↳ Basically applies additional row exchanges.

$$\rightarrow P^{-1} = P^T$$

• How many 4×4 Permutation matrices? $\Rightarrow 24$ P's.

Linear Algebra 18.6 - Lecture 5

Section 2.7 Section 3.1

Transposes, Permutations, Vector Spaces

Permutations P : execute row exchanges.

$$A = LU = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 0 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} \diagup & & & \\ & \diagdown & & \\ & & 0 & \\ & & & \diagdown \end{bmatrix} \text{ exchanges}$$

becomes

$$\hookrightarrow \underline{PA = LU} \quad // \text{ any invertible } A //$$

Permutations:

P = identity matrix with reordered rows.

$$\hookrightarrow (n!) = n(n-1) \dots (3)(2)(1)$$

$\hookrightarrow$ counts reorderings

$\hookrightarrow$ counts all $n \times n$ permutations.

$$(4 \times 4) \text{ matrix} \Rightarrow n! = 4! = 24$$

permutation matrices

$$P^{-1} = P^T \quad (P^T P = I)$$

Transpose

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\text{Transpose} \quad (A^T)_{ij} = A_{ji}$$

Symmetric Matrices $A^T = A$

e.g. $\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 9 \\ 7 & 9 & 4 \end{bmatrix} \rightarrow$ Unchanged by transposing the matrix.

e.g. 2: $R^T R$ is always symmetric

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^T \cdot \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 11 & 7 \\ 11 & 14 & 10 \\ 7 & 10 & 18 \end{bmatrix}$$

How do I know? Why is it symmetric?

$$(R^T R)^T = R^T \cdot R^{TT} = R^T \cdot R$$

$$\therefore (R^T R)^T = R^T \cdot R$$

• Vector Spaces

- ↳ What are Vector Spaces?
- ↳ What are Subspaces?

Examples: $\mathbb{R}^2$ = all 2-dimensional real vectors = "x-y plane"

$$\dots \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \pi \\ e \end{bmatrix}, \dots$$

Add $\begin{bmatrix} 3 \\ 2 \end{bmatrix} + \begin{bmatrix} \pi \\ e \end{bmatrix}$

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

- Vector space $\rightarrow$ all linear combinations remain in same plane.
↳ {own note}

- $\mathbb{R}^n$ = all column vectors with real components.

$$\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$$

↳ Adding & multiplying results in vector in the same plane.

↳ Note: 8 Rules define operations on these vectors $\rightarrow$ in textbook.

- Not a Vector Space

↳ - Vector space?

• Subspaces

- a vector space inside $\mathbb{R}^2$
- subspace ~~inside~~ of $\mathbb{R}^2$
- Line in $\mathbb{R}^2$ through zero vector
- Dotted line in the subspace?

Possible Subspaces of $\mathbb{R}^2$.

- ① all of $\mathbb{R}^2$
- ② any line through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.
- ③ zero vector only, $\mathbb{Z}$, $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

For $\mathbb{R}^3$

- ↳ plane through origin
- ↳ line through origin
- ↳ itself
- ↳ origin.

• Subspaces con.

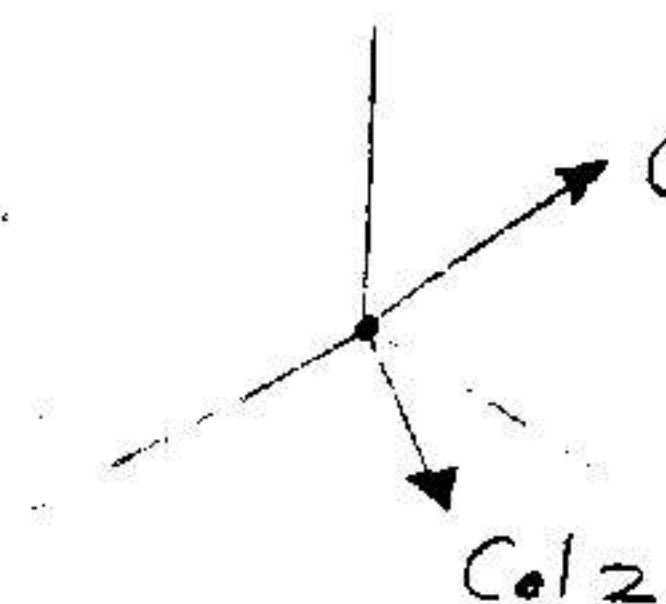
$$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$$

Columns in $\mathbb{R}^3$

All linear combinations form a subspace

↳ Multiply by something and add.

↳ Called column space $C(A)$ "C" of "A"



→ Take combination of
all $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}$ to form
a plane in $\mathbb{R}^3$

↳ plane goes through the origin

• How to create a subspace? (from a matrix)

↳ Take columns

↳ Take their linear combinations

↳ Get a column space $C(A)$.

• Linear Algebra 18.06 - Lecture 6

• Vector Spaces and Subspaces

- ↳ Column space of A : Solving $Ax = b$
- ↳ Nullspace of A .

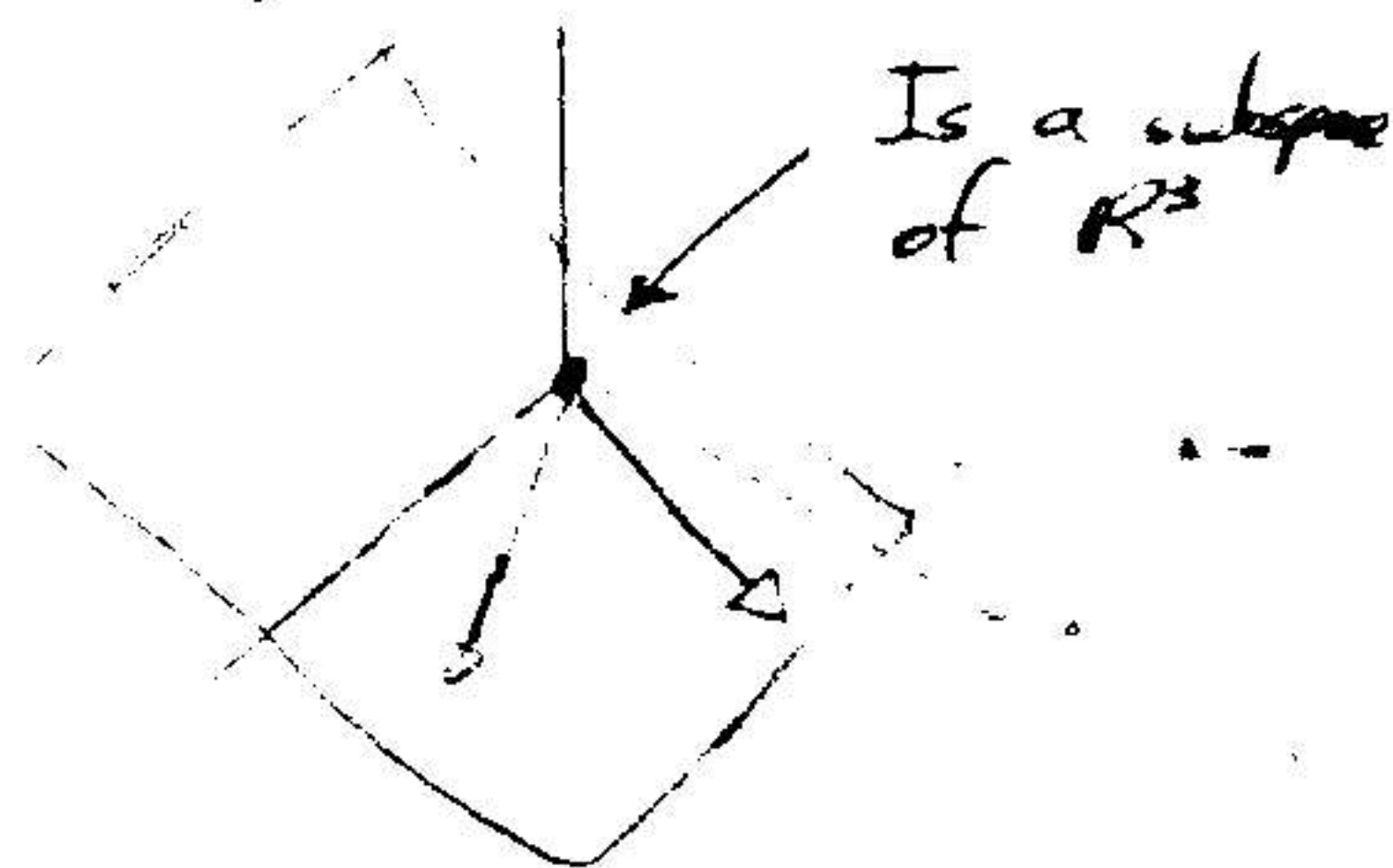
• Vector Spaces

Vector Space Requirements

- $\underline{v + w}$ and $\underline{cv}$ are in the space
- All linear combinations $\underline{cv + dw}$ are in the space

• Subspaces

- A vector space inside a vector space
- Plane through $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$



- ↳ 2 Subspaces: P and L

$P \cup L$ = all vectors in P or L
or This (is) (is not) a Subspace or both.

$P \cap L$ = all vectors in both P and L .

- Subspaces S and T intersection $S \cap T$ is a subspace
 - ↳ Vectors both in S and in T simultaneously
 - ↳ Whereas $S \cup T$ vectors could be in S but not T or vice versa.

• Column Space of a Matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

- ↳ is a subspace of $\mathbb{R}^4$
- ↳ because A is 3×4 .

Column space of A = all linear combination of columns

Does $Ax = b$ have a solution for every b ?

↳ No.

↳ 4 equations ; 3 unknowns.

$$Ax = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ so } b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Cannot always solve 4 eqs with 3 unknowns.
↳ Put something in S you can't

e.g. $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 4 \end{bmatrix}$ we can solve $y = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Can solve $Ax=b$ exactly when b is in the Column Space.

↳ Column space contains all combinations.
↳ So has α .

Can I throw away any column and have the same Column Space?

↳ Yes. Column 3 $\rightarrow$ Col ① + Col ② = Col ③
Could do with Col ① as well.

↳ Then we would have a: 2 dimensional Subspace of R^4

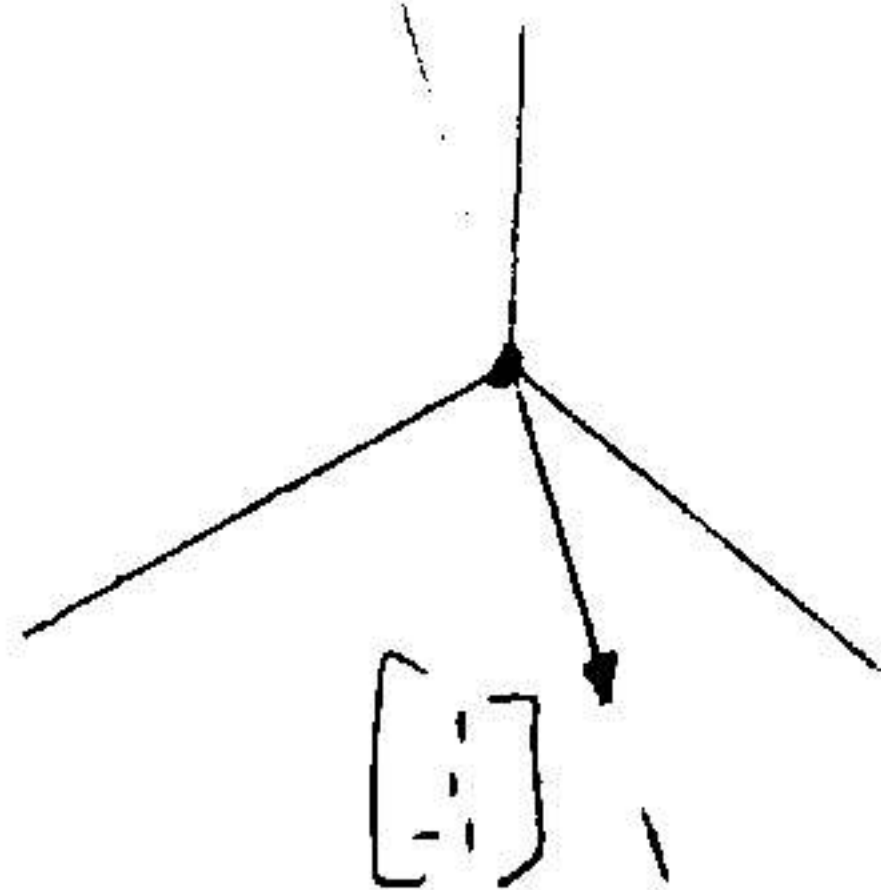
• Nullspace

Nullspace of A : $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$ $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ $\left. \vphantom{\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}} \right\} C(A) \text{ in } R^4$

Nullspace of $A \rightarrow$ all solutions $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ to $Ax = 0$

Nullspace $N(A)$ contains $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$, etc. $\hookrightarrow$ in R^3

↳ $\begin{bmatrix} c \\ c \\ -c \end{bmatrix} \rightarrow c \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ which is a line in R^3



• Check that Solutions to $Ax = 0$ always gives a Subspace.

$$\left\{ \begin{array}{l} \text{If } Ax = 0 \text{ and } Aw = 0 \text{ then } \underline{A(v+w) = 0} \\ \text{then } \underline{A(2v) = 0} \Rightarrow \boxed{2Av} \end{array} \right\}$$

Could have $Ax = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ but $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ is not a subspace
It forms a line (or plane) that does not go through the origin.

Linear Algebra 18.6 - Textbook Notes - Lectures 4-6

2.6 Elimination = Factorization: $A = LU$

Key Points

1. Each elimination step E_{ij} is inverted by L_{ij}
↳ Off the main diagonal change $-L_{ij}$ to $+L_{ij}$
2. Forward elimination process is inverted by L .
• $L = (L_{21} L_{31} \dots L_{n1})(L_{32} \dots L_{n2})(L_{43} \dots L_{nn}) \dots (L_{nn,n-1})$
3. L is lower triangular. Multiplier L_{ij} is in Row i Column j
4. Original A recovered from U by: $A = L \cdot U = \text{lower} \cdot \text{upper triangular}$
5. Elimination on $Ax = b$ reaches $Ux = c$
6. Solving a triangular system $\Rightarrow \frac{n^2}{2}$ multiply - subtracts.
↳ Elimination to find U takes $\frac{1}{3}n^3$

• Note $\rightarrow$ assume no row exchanges $\rightarrow$ predict L and U

When a row of A starts with zeroes, so does that row of L .
When a column of A starts with zeroes, so does that column of L .

Alternatively

• Triangular Factorisation $A = LU$ or $A = LDU$

↳ When seeing LDU , it is understood U has 1's on the diagonal.

↳ Each row is divided by its first nonzero entry - the pivot

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 8 \\ 0 & 5 \end{bmatrix} \text{ splits into } \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

1 Square System = 2 Triangular Systems

↳ How do we solve $Ax = b$?

1. Factor (into L and U , by elimination on left matrix A)
2. Solve (forward elimination on b using L , then back substitution for x using U .)

Example 3

$$\boxed{Ax = b}$$

$$\begin{aligned} u + 2v &= 5 \\ 4u + 9v &= 21 \end{aligned}$$

becomes

$$u + 2v = 5$$

$$v = 1$$

$$\boxed{Ux = c}$$

{elimination}

→ Multiplier was 4, which is saved in $\underline{L}$. Right side used 4 to change 21 to 1.

$$Lc = b \rightarrow \text{Lower triangular system} \rightarrow \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} c \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 21 \end{bmatrix} \Rightarrow c = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$Ux = c \rightarrow \text{Upper triangular system} \rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ x \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \Rightarrow x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

→ $\underline{L}$ and $\underline{U}$ can go into n^2 storage locations which originally held $\underline{A}$ (now forgettable).

• 2.7 Transposes and Permutations

1. Transposes of $\underline{Ax}$ and $\underline{AB}$ and $\underline{A^{-1}}$ are:

$$\begin{aligned} &= x^T A^T \\ &= B^T A^T \\ &= (A^T)^{-1} \end{aligned}$$

2. Dot product is: $x \cdot y = x^T y$. This is $(1 \times n)(n \times 1) = (1 \times 1)$

3. Idea behind A^T : $(Ax)^T y = x^T A^T y = x^T (A^T y)$

4. Symmetric matrices: $S = S^T$ (product $A^T A$ always symmetric)

5. Orthogonal matrices: $Q^T = Q^{-1}$ (Columns of Q are orthogonal unit vectors).

6. Permutation Matrix P has same rows as I in any order with $n!$ orders

7. $\Rightarrow P x$ puts components $x_1, x_2, x_3, \dots, x_n$ in that new order

$$\hookrightarrow \text{And } P^T = P^{-1}$$

• Transpose

→ Transpose : $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$ then $A^T = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 4 \end{bmatrix}$

→ Exchange Rows and Columns $(A^T)_{ij} = A_{ji}$

→ Transposing A, B :

together

$$(A + B) \rightarrow (A + B)^T$$

separately

$$A + B \rightarrow A^T + B^T$$

Sto Sum

$$A + B \text{ is } A^T + B^T$$

Product

$$(\cancel{AB})^T AB \text{ is } (AB)^T = A^T B^T$$

Inverse

$$A^{-1} \text{ is } (A^{-1})^T = (A^T)^{-1}$$

$$\begin{aligned} \hookrightarrow A^{-1}A &= I \\ A^T(A^{-1})^T &= I \end{aligned} \quad \square$$

• Inner & Dot Products

$x \cdot y$ is instead (using matrix notation) $x^T y$

T is inside Dot or Inner product is $x^T y$ $(1 \times n)(n \times 1)$

T is outside Dot or ~~Inner~~ ^{Outer} product is xy^T $(n \times 1)(1 \times n)$

$\hookrightarrow x^T y$ is a number

$\hookrightarrow xy^T$ is a matrix

• Symmetric Matrices

$$S = S^T, \text{ which means } S_{ji} = S_{ij}$$

→ transpose of $A^T A$ is $A^T (A^T)^T$ is $A^T A$

→ If $S = S^T$ is factored into LDU (with no row exchanges), then U is L^T $\therefore S = L D L^T$

• Permutations (accounting for row exchanges)
When to complete elimination,

1. Row exchanges done in advance ∴ $PA = LU$
2. Row exchanges done after elimination ∴ $A = L.P.U$

→ If A is invertible, a permutation P will put its rows in the correct order to factor $PA = LU$.

↳ Must be full set of pivots after row exchange

• 3.1 Spaces of Vectors pg 123.

1. Standard n -dimensional space $\mathbb{R}^n$ contains all real column vectors with n components.
2. If $\underline{v}$ and $\underline{w}$ are in vector space S , every combination, $c\underline{v} + d\underline{w}$ must be in S .
3. "Vectors" in S can be matrices or functions of x .
 - ↳ 1 point space Z consists of $x=0$.
4. A Subspace of $\mathbb{R}^n \rightarrow$ vector space in $\mathbb{R}^n$
 - ↳ vector space within a vector space.
5. Column space of A contains all combinations of columns of A
 - ↳ $\therefore$ a subspace of $\mathbb{R}^m$
6. Column space contains all vectors $A\underline{x}$
 - ↳ So $A\underline{x} = \underline{b}$ solvable when $\underline{b}$ is in $C(A)$.

$\rightarrow$ Essential Vector Operations

↳ go on inside the vector space, and they produce linear combinations

1. We can add any vectors in $\mathbb{R}^n$,
2. and we can multiply any vector by any scalar c .

"inside vector space" $\rightarrow$ result stays within the ~~vector~~ space.

$\rightarrow$ Other Vector Spaces

$M \rightarrow$ vector space of all real 2×2 matrices

$F \rightarrow \dots$ all real functions $f(x)$

$Z \rightarrow \dots$ that consists only of a zero vector.

• 3.2 Nullspace of A

1. Nullspace $N(A)$ in $\mathbb{R}^n$ contains all solutions x to $Ax=0$
2. Elimination does not change the nullspace
3. The reduced row echelon form $R = \text{ref}(A)$ has all pivots $= 1$, with zeros above and below.
4. If column j of R is free (no pivot), there is a "special" solution to $Ax=0$ with $\underline{x_j = 1}$
5. Rank $\rightarrow$ Number of pivots $=$ number of nonzero rows in R
 $= \text{rank } r$ ($n-r$ free columns)
6. Every matrix with $m < n$ has nonzero solutions to $Ax=0$ in its nullspace.

- "Special" Solution

$As=0$ ~~$N(A)=$~~
 Nullspace of $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ contains all multiples of $s = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$
 $\hookrightarrow$ Solution is special because we set free variable $x_2 = 1$

- Process Steps

- (1) Reduce A to its row echelon form R
- (2) finding special solutions to $Ax=0$.

• Pivot Columns and Free Columns

~~First column of $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ contains the only pivot, so~~
 Free Components $\rightarrow$ correspond to columns with no pivots

• Reduced Row Echelon Form

When A is rectangular, elimination will not stop at the upper triangular U .

$\hookrightarrow$ Matrix can be made simpler: (brings us to best matrix R)

- ① Produce zero above the pivots $\rightarrow$ Use pivot rows to eliminate upward in R .
 $\begin{bmatrix} \square & 0 & 0 \\ \square & \square & 0 \\ \square & \square & \square \end{bmatrix}$
- ② Produce ones in the pivots $\rightarrow$ Divide whole pivot row by its pivot.
 $\begin{bmatrix} \square & 0 & 0 \\ \square & \square & 0 \\ \square & \square & \square \end{bmatrix}$